

Aspirin inhibits the enzyme cyclooxygenase. Researchers discovered a new drug that inhibits cyclooxygenase more effectively. If they report that the difference between aspirin and the new drug is statistically significant, it can be assumed that they:

- ☐ A. rejected a true null hypothesis as there was no real difference between the groups. [9%]
- ✔ ☒ B. rejected a false null hypothesis as there was a real difference between the groups. [71%]
- ☐ C. failed to reject a false null hypothesis as there was no real difference between the groups. [2%]
- ☐ D. failed to reject a true null hypothesis as there was a real difference between the groups. [16%]

Correct

70% answered correctly

Explanation:

		H ₀	
		True	False
Decision about H ₀	Reject (p-values ≤ α)	Type I error (Falsely conclude there is a true difference)	Correctly conclude there is a true difference
	Fail to Reject (p-values > α)	Correctly conclude there is no difference	Type II error (Falsely conclude there is no difference)

H₀ = null hypothesis that there is no true difference between 2 groups
 α = significance level, often set to 0.05
 p-values ≤ α are considered statistically significant

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A hypothesis test is performed to assess the validity of a claim made about 2 groups. The **null hypothesis (H₀)** theorizes that there is no difference between these 2 groups. In contrast, the **alternative hypothesis (H_A)** rivals the null by claiming that a difference does exist. *p*-values evaluate how well the data support H₀, assuming H₀ is true. The statistical significance of the results is determined by whether the ***p*-value** is equal to or less than α, the **predetermined level of significance**. In general, α is set to equal 0.05, which means that ***p*-values ≤ 0.05** are considered **statistically significant**.

The researchers found that the new drug they discovered inhibits cyclooxygenase more effectively than aspirin does. To arrive at this result, they first established the following hypotheses:

- H₀: The inhibition of cyclooxygenase by the new drug will not differ from that of aspirin.
- H_A: The inhibition of cyclooxygenase by the new drug will differ from that of aspirin.

The researchers' claim of statistical significance means that the resultant *p*-value they obtained was ≤ α. Therefore, H₀ is false. They rejected a false H₀ as there was a true

difference between the groups (H_A).

(Choice A) Rejecting a true null hypothesis (type I error) leads to the incorrect conclusion that there is a difference between groups. In reality, there is no difference between groups.

(Choice C) Failing to reject a false null hypothesis (type II error) leads to the erroneous conclusion that there is no difference between groups. In reality, there is a difference between groups.

(Choice D) Failing to reject a true null hypothesis leads to the correct conclusion that there is no difference between groups. However, the null hypothesis is false in this case and was rejected as a result.

Educational objective:

The null hypothesis (H_0) theorizes that there is no difference between 2 groups, but the alternative hypothesis (H_A) rivals the null by supposing that a difference does exist. The significance of the results is then determined by whether a p -value is equal to or less than α , the predetermined level of significance.

Time spent:0 sec

Subject:
Biochemistry

QID:402107

System:
Enzymes